

OBSERVATION

UNTIL the end of the previous chapter, we had been taking observation and the idea of observational evidence pretty much for granted. Some scientific questions, we have assumed, can be answered just by looking. Even if the logical implications about the truth of theory are unclear in the process of testing, the outcomes of the tests, whether the predictions hold true or not, have been taken as unproblematic. The test results, the observations, are of a purely manifest world of appearance. They are there for anyone who takes the time to look, and they can be used by anyone as impartial evidence. When in doubt about theory, there is always **observation**.

Observation is the touchstone of external virtues. Empirical adequacy, that is, compatibility with observation, is presumably easy and uncomplicated to evaluate. And observation is the distinction of external virtues, since it is the contact of theory to the outside world. Observation is the source of the world's input and guidance over our describing and modeling. We have separated external from internal virtues under a presumption that the world's input through observation puts the external features beyond our control in the sense that it's not possible to force the observations to come out one way or another. They are forced on us and we must take them as given. Observation is, if anything is, a source of objectivity in science.

This has been the expectation of the activity of observation that has motivated the internal–external distinction and has been quietly at work in the analysis of explanation and confirmation. It is an expectation that deserves a closer look, given the critical

role observation is to play in the scientific process of justification. Objectivity, after all, is no insignificant burden. And in light of the complication suggested in Chapter 5, the notion of degrees of observability parallel to degrees of indirectness in the observation, the nature of scientific observation warrants a careful look. The concept of observation is used rather liberally by scientists, to judge by the sorts of things they claim to have observed. Protons have been observed, insofar as their distinguishing features, such as mass, electrical charge, and spin, have all been measured. And of the six varieties of quarks, it is not unusual to hear that five have been observed. The search continues for the one called the top quark. (Interestingly, it is also known as the truth quark. The search continues for the truth. Up, down, strangeness, charm, and beauty are all in, but truth is a continuing holdout.) The new Hubble telescope now in orbit around the earth is supposed to be so powerful and unbothered by atmospheric distortion that it should facilitate the observation of objects so distant, receiving light that they emitted so long ago, that we could see these objects as they were at very nearly the beginning of time. We can observe the big bang with this thing.

This may be how scientists invoke the notion of observation, but are these examples of information straight from nature, free of any conceptual embellishment, the undeniable, impartial, common ground of objectivity? And if these examples do not play that role, are there any scientific activities that do? Is the pure observation that has been taken for granted a real possibility?

THE IMPORTANCE OF OBSERVATION

There should be no doubt that the activity of observation is an essential aspect of the scientific process, but it is wise to emphasize this so that the importance of observation does not get lost in

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the following analysis. Asking questions of the activity and demanding an account of the source and handling of information should not be interpreted as an attack or a disguised criticism. Observation, like science in general, is scrutinized because it is so important, not because we expect to find flaws.

The sciences under account here are one and all empirical sciences. Whether they are natural sciences like physics, biology, and geology or social sciences like psychology and anthropology, the claims they make are contingent on what is going on in the world. Whether it is the world of people or of elements, organisms, and the environment, it is an actual world that is to be described by science. Contrast this with claims made by nonempirical enterprises such as mathematics or logic. These describe how some of the basic relations among things *must* be; the truth of these claims is not contingent on the way things in fact *happen* to be, and the truth of logical and mathematical propositions can be assessed without having to look out at what's happening in the world. It is a logical truth that outside at the moment it is either snowing or it is not snowing. I don't have to look outside to know that this is true. It *must* be true. This difference between necessary truths and claims whose truth is contingent on the way things happen to be shows up in the process of reading a book, as well as in science and daily life. It is the difference between descriptions of how books in general must be written, rules of grammar and noncontradiction, for example, and descriptions of how in fact *this* book that I am reading is written and what the writing says. The only way to tell what this particular book says is to read it, just as the only way to tell if it's snowing at the moment is to have a look.

Reading the book of nature, our metaphor for doing natural science, requires observing the events in nature. To have any claim to an accurate report of what's going on in *this* world, there must be input from the world. Observation, in other words, is a dietary staple of empirical science.

As the channel of information from the world, a great deal of expectation is placed on observation to be the decisive and authoritative arbiter of truth in science, and it is exactly this hopeful role that we are looking into. Empiricists in particular might hope that the reports of observation will serve as a common ground of agreement for all participants of science regardless of their theoretical or skeptical preconceptions. Observations, being imposition from the world, are intersubjective and public in a way that beliefs and theories don't have to be. They are accessible by everyone and, the hope continues, agreeable to everyone. As a common measuring stick, observations can be appealed to as adjudicators of theories to convince skeptics and those of alternative theoretical persuasion. When all else fails, in other words, and all manner of persuasion is inconclusive, we can at least all agree on the observations. They are the terminus of dispute.

This expectation of neutrality derives from a presumption that there will be some basic form of observation that is free of interpretive influence. The usual contrast is between observation and inference where the latter is the general description of the human contribution. When inferences and interpretations are involved there is the danger of error and bias, and the products of thought could be mere artifacts with no correlation to the real world. Observation, in contrast, is hoped to be just the facts, the unaltered information from the world.

There is the hope of the empiricist, and intuitively there is plenty of plausibility with the idea that there are some observational claims that any reasonable, unimpaired person will agree to. To strengthen the pull of this intuition, suppose we are given a questionnaire of true-false questions about items in the room. Half the questions are about the books in the room, stating such things as: (1) More than half the books in the room are blue. (2) None of the books in the room is on the floor. (3) There is at least one book in the room. And so on. The other half of the questions concern the cosmic rays in the room, and the statements to

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be judged true or false here are quite similar to those about books: (1) There is at least one cosmic ray in the room at any one moment. (2) On average there are more cosmic rays going down than there are going horizontally. (3) There are fewer cosmic rays in the room on Sunday than on any other day of the week. Making responsible decisions as to which claims are true and focusing on the features of a claim that indicate that it is likely to be true are exactly the tasks of justification. Science is like a true-false exam as we must responsibly decide, among proposed theories, which to endorse as true.

The telling factor in this questionnaire is not which of the statements are true or which ones can be answered. The important issue is which of the answers can be checked and agreed upon. With the books in the room, this checking and agreement come easily. Just have a look. Count the books, note their color, check the floor. Anybody can do this, and anyone who does and who is reasonably participating in a search for the truth will agree to the right answer. Books are observable. Cosmic rays are a different story. To check the answers to the true-false questions about cosmic rays we must either consult the relevant theories, those that describe what cosmic rays are, where they are likely to be, and what they are likely to be doing, or use a device such as a cloud chamber to detect the cosmic rays and reveal their activities. A properly prepared and oriented cloud chamber may reveal transient streaks of vapor, which are the telltale tracks of cosmic rays. As with the case of books, now all we have to do is have a look, point to the chamber, and say, "There they are! There are cosmic rays in the room." The similarity to observing books is shaken, though, with the question of why anyone should believe that these are in fact cosmic rays, in other words, when we question the responsibility behind the endorsement of claims about cosmic rays. Why do we agree with the report "There are cosmic rays," as we must if it is to function as a common ground of evidence? The answer is in theory, the theoretical account of the

formation of the streaks, citing, as it does, cosmic rays as the efficient cause of the visible tracks. But the agreement, the common ground, disappears as long as there is an alternative theory of the formation of the lines. If someone suggests, for example, that the lines are formed by a magnetic effect and that what the cloud chamber shows are actually the magnetic field lines of the earth and nearby magnets, the claim about cosmic rays is not decidable without thought and the human contribution of interpretation. We are at the mercy of theory to check the answers about cosmic rays. But no one can pull any funny stuff about books. Whatever they claim about the books, we can just have a look for ourselves.

The empiricist's hope for observation, and the motivation for distinguishing observables (like the books and the streaks of vapor) from unobservables (like cosmic rays), is to escape from being at the mercy of theory. The hope is to identify some data that we can check for ourselves, information that is there for the taking because it is given by nature.

This tale of cosmic rays points out that the focus of our understanding of observation must be on the observational report rather than on the physical event of sensation. It is crucial to pay attention to what is claimed to be seen. The same physical event of looking at the cloud chamber could be reported as observing cosmic rays or as observing streaks. The difference in justification of the two reports is an important difference. We must attend to observational reports to see whether it is possible to fulfill the hope of escaping from the mercy of theory.

ACCOUNTABLE OBSERVATION

We are interested in events of observation that can participate in the process of science. To be useful as evidence an observation must make contact with a theory in some way if it is to serve as proof or as refutation. Science can make use of an observation

only if it is an accountable observation, and this accountability has two aspects: The observation must be informative in the sense that it must be an account *of* something if it is to have the assertive content required to serve as justification of other claims. And the observation itself must be justified in the sense of being certifiably not haphazard or uncontrolled.

These two features of accountability are required of observations if they are to be informative and useful contributors to knowledge. To be evidence for a theory, an observation must be evidence *of* something. The proper question to think of observation as responding to is not simply "Do you see it?" but "*What* do you see?" or "Do you see that this is a *that*?" Accountable observation is more than just opening your eyes and saying "Yep, I see it." It requires a claim as to *what* is observed and an assertion of the *information* that is being conveyed by the world. One needs to know what it is evidence of if the results of observation are to be related to other ideas and compared to theoretical predictions.

Again, the lesson is to focus on the observational report. Even if the report is private, a report to oneself, the point is that the sensation must be described as a *this* or a *that* if it is to be useful. Simple observations such as the ones used to check answers to the questions about books in the room must be rendered in a conceptual description if they are to be compared to the concepts of the questions they are checking. A useful observation must be reported in an informational form, observing, for example, *that* the book is on the floor. Undescribed sensations are useless as evidence. Or perhaps it is better to say that pre-description sensations are pre-useful. Objects of thought require conceptual handles, and there is no way to think about ideas and to relate them to other ideas without the conceptual organization that comes with description. And if an observation report is to be relevant to a theory it must include some of the language of theory. It must assertively be evidence of an *x* if it is to be evidence in the case to justify a theory about *x*. To serve as justification, an

observation must have some assertive content, and this is half of the requirement of accountability.

The other half is that the observation report must itself be justified if it is to have sufficient assertive authority to prove other claims. Science cannot accept just any old report of the senses. As a rule, observations must be carefully done and repeatable. They must be carried out in a controlled way and under the proper conditions. Furthermore, and this is important, one must be aware of the relevant conditions and the factors of care, repeatability, and control. It's not enough to insist on observations that *are* accurate or very nearly accurate. We must insist further that one *can tell* when they are accurate. We must insist, that is, on verifiable observations. We don't want to rely on dumb luck that the observation reports are accurate. Responsible observing is observing that we are able to justify. This requires an understanding of the relevant conditions under which an observation is done. It does not require that each individual observer understand these conditions, but the scientific community must understand what it takes to do the observation properly if the observation is to be acceptable as scientific evidence. Science, after all, is a public enterprise, and matters of justification are matters of community.

It is not that this verification report in fact accompanies every scientific observation, nor should it. The point is only that such verification is available. An acceptable observation report is one for which justification in terms of an account of the proper conditions and procedures *can* be given. In science, an observation cannot be entered as evidence unless it can be accredited in this way.

Thus are the two edges of accountability. To contribute to science or to knowledge in general, an observation must be about something. That is, it must be informative, assertive, propositional. And there must be some reason to think that the observational report is accurately about what it claims to be about. There is nothing absurd in the notion of an inaccurate observa-

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tion, and so there is nothing out of line in the concern to block false observation reports from being entered as evidence. Beyond the immediately perceptual distortions such as dreams and hallucinations, there are observational distortions that result from unsuitable conditions. The observer may be improperly attentive or too far away. The viewing conditions may be too dark or littered with obstructions. Or there may be interference from outside sources causing a distorted view. It may all have been done with mirrors. In science, as in any responsible attempt to know about the world, we must be prepared to convince ourselves and others that none of these things has gone wrong and that what we claim to observe is an accurate report.

Recognizing the need for accountability of observation points out that observational information is not simply given to us by the world. Observation is not a matter of passive reception of ready-made, in the sense of ready-described and ready-justified, facts. Observation, insofar as it is a useful constituent of knowledge, is an activity of description and justification.

These are the general requirements of useful observation. The question now is, How are these requirements met in science?

OBSERVATION AND THEORY

The requirement that observation is to contribute to scientific knowledge, that is, that what is considered as an observation must be a knowledge event and not just a psychological event, has an important consequence: All observation in science is influenced by theory. This is not simply the way things happen to be done; it is the way things have to be.

Tying all observation to theory is a sweeping and a vague proposition, and it is of pivotal importance to the process of justification in science. It is therefore crucial to see the details of just how theory influences observation and how this affects the role of observation as objective adjudicator of the truth of theories. The

suggestion is that what is observed in the world, that is, what is reported from observation, depends on the background theories held by the observer. Different theoretical backgrounds, as between someone advocating the caloric theory of heat and someone else believing the kinetic theory of heat, will lead to different observations in the pool of acceptable evidence. In other words, informative observation is impossible without a theoretical background, and alternative theoretical backgrounds lead to alternative informational reports. To paraphrase Kant, observation without theory is blind.

Enough with the slogans. It's time for details.

Theory influences observation in several ways. For one, our theoretical understanding of the world guides our selection of what observations to do. We cannot look at everything. Scientists must be selective of a few observations worth their time and money. The business of gathering data is neither comprehensive nor random, so it must be directed by theories that tell us where to look and what is important enough to warrant attention. Einstein's theory indicated that an observation of stars during an eclipse would be informative. Solar theory directs the observational efforts toward a search for neutrinos. This is of course the way things have to be and ought to be if the idea is to test a theory by its consequences. The point is simply to make plain the situation that the theories in place direct the selection of evidence. Different theories would precipitate a different collection of evidence. So when we talk about comparing a theory to the observational information from the world, the situation is more accurately described as comparison to a very small sample of information from the world, a theoretically specified sample. The sampling is done with an attention to relevant factors. That is, irrelevant variations need not be included. If we drop a brick and a sponge to test the parity of gravitational acceleration, we don't have to run the test with a sponge bought at Woolworth's and again with a sponge from Sears, or once with a blue one and

again with a pink one. Where you buy the objects, certainly, and their color, probably, have no relevance to their gravitational or inertial properties, so these observations need not be made. Their exclusion from the collection of evidence is based on an understanding of the causal mechanisms involved, and that is a theoretical understanding. The relevance of an observation, and hence what is chosen to be observed, is influenced by the theories we believe.

A second way that theories influence observation is by guiding decisions as to when observations are credible. Theories play a key role in the accountability of observations by supplying standards to evaluate the reliability of observational reports. The assessment of the viewing conditions, attesting that there are no distorting factors or correcting for any distortions that persist, is based largely on an understanding of the causal mechanisms of observation. Clearly, there must be adequate lighting if a visual observation is to be credible. Why? Because vision is accomplished by receiving the light energy that bounces off the specimen and is transmitted through the space to the observer and into the eye. Light carries the information. Furthermore, it is no threat to the accuracy of the observation if the light must travel through air before reaching the observer. A theoretical understanding of the optical properties of air assures that, unless there is significantly uneven heating, light will pass straight through. And of course a well-made windowpane is no distortion to observation. It's not that each of us has to know a lot of theoretical optics to be able to look outside or to claim that it's snowing. It is rather that, if we are questioned on the legitimacy of the observation, there are these sorts of theoretical accounts to accredit the accuracy of the report. This sort of accountability is required of scientific observation, and it is clear that a different understanding of the causal interaction, that is, a different theory of what's going on in the event of observation, may sanction different observation reports as acceptable.

This role of theory is especially apparent in cases of observations that are assisted by machines. This is a very large class of important scientific observations. Telescopes, for example, if properly constructed, do not manufacture false images. They faithfully reproduce what is out there beyond the visual reach of the eye. In the case of observing the stars as a way to test the general theory of relativity, it wasn't the telescope that altered the apparent positions of the stars; it was a real effect of the sun. The telescope only magnified the view. Any distortions or alterations of the specimen can be accounted for and even corrected because we know how the machine works. The light from the stars is bent and focused by mirrors and lenses to produce an image with all of the relevant features of reality. This trust of accuracy is sanctioned by a theory of optics. Similar theories are the basis of our belief in the accuracy of images produced by a microscope. Reports about the microscopic specimen are not acceptable without a theoretical account of how the imaging is done. Such reports are not required with every reported observation, but they must be available. They are the answer to the question, Why should we trust the images produced by this thing? – should that question ever come up.

One doesn't have to understand the theory of how an instrument such as a microscope or a cloud chamber works in order to use the instrument. Noticing amazing things like the teeming life in a drop of pond water does not require an awareness of optical theory. You or I or anyone can use a microscope to see the single-celled creatures. Is this observation without theory? Biologists use microscopes routinely without really knowing how they work. And relatively unskilled and theoretically illiterate lab technicians are often told what to look for in the apparatus, and they succeed in observing what they should even with no theory in mind to influence their actions. In the good old days, the many photographs from particle detectors were

scanned by hourly workers without a glimmer of understanding of particle physics. They were trained only to notice certain branching patterns in the streaks on the photograph and to note their frequency. Is this observation without theoretical influence?

Certainly not. A great amount of theory goes into the building of the machines, the particle detectors and microscopes, and into sanctioning their reliability. The data from these machines are acceptable, that is, the observations are accountable, only if the experts, those who invent and build the machines, have a solid theoretical account of how they work. Machine-aided observation is a communal event. Anyone can notice interesting features of the image, but the process is accountable only in light of the theories that describe how the image is formed. It is this accountability that gives the observation its license to contribute to knowledge.

It is useful to refer to these theories that describe the chain of interaction from the specimen to the observer as the **accounting theories**. They are not special kinds of theories with a special content. They are regular old theories with a particular use. Any theory could be used as an accounting theory. In cases of unaided vision, for example, basic theories of optics play the role of accounting. They answer for the accountability of the observation. With observing apparatus, the interaction chain is longer, as is the list of accounting theories. More theoretical claims will usually be invoked to verify that the image does indeed have information of the specimen.

The accounting theories, in other words, help clarify what the observations mean, in the sense that they specify what the images are images of. Accounting theories guide the appropriate description of the observation. Looking at the cloud chamber, for example, one is licensed to report that we see more than just streaks; we see images of cosmic rays, and the license is issued under the

authority of a theoretical account of the cosmic rays, being electrically charged particles, ionizing the gas in the chamber, and thereby causing visible condensation along their flight path. The accounting theories fit the manifest image, the streaks in the cloud chamber, into the descriptive language of theory. This is a necessary step in the process of testing. Cosmic-ray theory will generate predictions about cosmic rays, and to test the theory these predictions must be imaged. Somewhere in the description of the observation, the test results must be about the objects of theory. This is how the observations are relevant to theory and can function as confirmation or refutation. Theory, in other words, guides the description of observation, lest the observation be irrelevant.

Even in cases where no apparatus assists the observation, describing what is observed has a significant theoretical influence. Simply calling something a this or a that requires having a concept of this or that. This application of concepts is done by grouping the individual items we encounter under headings of the same *kind* of thing. Things with relevant features in common are the same kind of thing. The individuals will not be exactly alike, but some features are irrelevant to membership in the category. Not all trees are green, nor do they all grow in Vermont, but these properties are not relevant to what it is to be a tree. Not all electrons are bound in atoms. There is a concept of being an electron that describes what it is to be that kind of thing, and that includes some individuals bound in atoms and some that are free. Specifying which features are relevant to the association of the kind of thing is a job for theory. Information on the grouping of items into kinds is not given by nature, since there are various ways to do it depending on which features are identified as important and relevant. This ruling on relevance is clearly influenced by theory, and in this way any activity of description, the application of a particular item to its conceptual kind, is guided by theory.

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To summarize this final manner of theoretical influence on observation, all useful data must be described data, and the link between sensation and descriptive concepts is a link made fast by theory.

THEORY-LADEN OBSERVATION

“Theory-laden observation” is the slogan often used to summarize the points made in the previous section. Observations in science are theory-laden. Accountable observation, the only kind that plays a contributing role in the acquisition and justification of knowledge about the world, bears an inseparable theoretical component. We are forced to give up the notion of pure, unaltered information from the world. Better to think of observational information not simply as given by nature but as actively taken by the observer in an activity that is guided by the theories, beliefs, and concepts in the mind of the actor. This dashes the empiricist’s hope of a common ground of evidence agreeable to everyone regardless of his or her theoretical predisposition. Different theoretical positions, perhaps rival theories that we are trying to decide between, could claim different evidential ground by authorizing different observations as important, different observations as reliably done, or even the same sensations differently described.

Accountable observation depends on accounting theories, whether or not the observation is done with the assistance of instrumentation. An observation’s authority to justify other claims, that is, its usefulness, is based on its assertive content and its accountability. The spontaneous, uncomplicated result of just looking at the world may be that we come truly to believe what we see. But that’s not good enough. To function as evidence in science, the act of observation must offer reason to think that we believe truly what we see.

The indelible theoretical component of observation means that observational beliefs are not rigidly foundational in the structure of justification. They are not immune to change or refutation. Observations carry relatively more authority than do more deeply theoretical claims in the sense that we are much more reluctant to deny the information of our eyes than information from more theoretical claims of things well removed from perception. As a rule, theories must bend to the mold of observation. But in the case of a single observation that conflicts with a large body of well-entrenched theory, it is an option wisely taken to discount the observation rather than theory. Better look again before chucking a healthy system of theories on the word of a single observation. It won't do to simply ignore the troublesome observation as if it never happened. Better to explain it, if possible, as a case of inappropriate conditions or, as a last resort, the distorting influence of some unknown interference. The point is that observations do not enjoy an absolute authority over theories. They too are acceptable insofar as they fit into the theoretical network. They are revisable under the influence of theories.

The picture then is not of observations dictating the acceptability of scientific claims. The relation between theory and observation is more of a republic where observations serve with authority by the will of the theories. As a populace chooses its rulers, theory vests observation with an authority to rule on the justification of theories. Observations in turn force theories to bend to their evidential rules, but they can force and bend only so far lest they lose their authority. Coherence among the theories, like consensus among the people, is still the ultimate source of persuasion.

The ambiguity inherent in the words "only so far" is exactly the same ambiguity in the process of falsification. In the event of an observation that conflicted with theory, a false prediction, one has to decide which theory, the hypothesis or an auxiliary, to reject. This is not a capricious decision. It is done with an eye toward maximizing the overall virtues of a whole network of

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theoretical claims. The current analysis of observation reports shows that they too are on the ballot for rejection or revision. One *could* reject the observation report. Again, the decision is guided by the participating theories and is done with a sensitivity to the plausibility of the accounting theories as well as the magnitude of disruption that would result if the observation is believed.

OBSERVING AND READING

The analogy between reading a text and doing science, used at the beginning of this chapter to accent the importance of observation, can serve as well to clarify the relation between observation and theory. In reading a book, we want to know the plot, the meaning of the text. The accomplishment of reading is to learn what is beyond the manifest marks on the page, that is, to understand what the marks indicate, what they mean. This of course cannot be done without carefully looking at the marks. So it is in science. The scientific accomplishment is to understand what the observations indicate, and this is done only through a careful attention to the observations themselves.

Marks on a page are not helpful at all unless they can be organized as letters, words, and whole statements. One step in the method of reading must be to form concepts on the units of meaning. So it is in science, where a methodological prerequisite to useful observing is to form concepts of things so that tokens of sensation can be grouped into interacting kinds of things and the relevant kinds can be individuated and understood.

Marks on a page, even when individuated and organized, are not helpful for understanding the text unless they are seen to indicate some piece of the plot. They must mean something if they are to be informative. This step from syntax to semantics is a huge one, as is all too clear to someone learning a new language. I can read modern Greek, for example, in the sense of being able to read the alphabet and pronounce the words. I even know

enough grammar to be able to identify verbs, nouns, adjectives, and so on, and I am a tireless annoyance to my Greek friends as I love to read out loud all the signs, menus, magazines, or whatever I see. But I haven't a clue what any of it means. Well, not much of it anyway.

The level of information I can get from a Greek text will not be sufficient for translating it into a meaningful message. It is not enough to allow me to understand the plot or to test out any speculations I might have about what is going on in a Greek book. The text is completely opaque without some claims of connection between the marks on the page and the meaning they convey. So it is in science, where observations are meaningful contributors to our understanding of nature only with some claims of connection between the image in sensation and the specimen in the world. To translate the book of nature one needs theories of what the observations mean, a lexicon of what the marks are evidence of.

BLURRING THE INTERNAL–EXTERNAL DISTINCTION

THE idea of distinguishing between the internal and the external features of theories and regarding these features separately as potential indicators of the truth of a theory was only a start-up suggestion. It has served as a heuristic organizer of the activities of science and the steps of justification. Such organizing concepts are always needed for thinking about evidence, evidence in this case of the methods of scientific justification and of the structure of scientific knowledge. It seemed natural to begin by dividing the fund of information that could be used to evaluate a theory into two categories, depending on the source of the information. There could be information from theories themselves, both the theory being evaluated and other theories relevant to it, and information from the world. Thus justification of a theory would be a combination of theory-to-theory relations, as in the standards of entrenchment and explanatory cooperation, and theory-to-world relations. This matched the internal–external distinction.

The initial suggestion that this is a sharp dichotomy, separating distinctive kinds of features, different sources of information, and characteristic manners of justification, has become untenable in light of the closer look at the relationship between theories and evidence in science. The developing model *of* science is proposed and tested just as a theory *in* science. Just as in science, an initial hypothesis was a helpful background against which to structure the evidence, but the hypothesis is answerable to the evidence and it must be revised or rejected if the evidence demands. In this case, the hypothesis about the structure of science is the claim of

dichotomy between internal and external virtues. The evidence it both helps to interpret and uses for testing is the detailed description of scientific theories and their justification. This dialectic between the model of science and the evidence about science, their influence on each other, indicates the need to revise the model at this point.

Consider the evidence. Considerations of external features such as a theory's facility at explaining phenomena and its record of successful prediction, are inconclusive without the complement of internal evaluations. What counts is being the *best* explanation, and this is measured by standards of fruitfulness and theoretical plausibility. Thus internal and external features of theories are inextricably mixed.

But more important than this mixture is the fact of science that the observations themselves bear a theoretical component. Each element in what has been regarded as an external test of theory, each appeal to observational evidence, is a comparison between hypothesis and other conceptually influenced beliefs. Observations, the evidential elements for justifying theories, are themselves assertive beliefs. They are caused by influences from the environment, but they are described and justified under the influence of other beliefs, that is, other assertive claims within our awareness. Observation reports are not so different from theories in that they acquire their conceptual relevance and their justification by their fit into the conceptual, theoretical system. Justifying a theory by comparing it to observation is an internal activity.

The complications that have been described in explanation, confirmation, and observation, the complications that have put justification into a situation of always comparing theory against theory rather than against facts in the world, are not avoidable complications. We point them out not to show what an awful mess science has gotten itself into and certainly not to advocate a way to sidestep the mess and get theory directly in touch with the facts. Rather, these factors of auxiliary hypotheses, underdetermi-

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nation, and the theoretical component of observation are inevitable. They are in the nature of any attempts to know more about the world than the manifest appearances. The important reason for pointing these things out is to make us aware of the predicament we are in. Only if the predicament is clearly understood can we figure out how to deal with it.

THE PREDICAMENT

The enterprise of justifying scientific theories is in many ways like taking a true–false exam. The result of deliberation over each theoretical claim is a decision whether to assent to it or not or, in a more refined way, a decision as to how likely it is that the claim is true. Each theoretical claim is a declarative assertion about the world, and the business of justification is the issue of truth. Sciences assert, for example, that the ocean’s tides are caused by gravity, and the process of justification is engaged to report on the warrant for believing this to be true. All sorts of hypotheses are proposed in science. Cosmic rays are the debris from reactions in distant stars. There are some cosmic rays in the room right now. Modern humans inhabited the North American continent as early as 15,000 years ago. All of these are true or false, and it is the burden of justification to decide which each is likely to be.

In science, no one ever gets a look at the answer key. There is no direct access to information on the truth or falsity of the theoretical claims made. This access problem is clear in cases of claims about unobservables or, respecting the degrees of observability, about things that are only distantly or indirectly observable. Things like cosmic rays, atoms, DNA molecules, or Greek Bronze Age palaces can never be plainly seen. There can be no clear view of such things with the unequivocal news that of this we have spoken the truth. There is at best evidence of these things, evidence, like lines in a cloud chamber or the half-buried ruins of stone and ceramic, that guides the assessment of

truth of the theoretical claims. The evidential claims themselves are accounted for by theories, showing that they too are but other entries on the exam and not inside information from the key.

Even evidential claims that are of observable things like the books in the room, the streaks in the cloud chamber, or the basic excavation data of fragmented ceramics and jumbled stones, even these claims are on the exam. These direct observations are not the answer key, the report on simply how things are in the world, for, as we have seen, they are subject to justification and are revisable under pressure from theories. The situation of coming to know about the world doesn't allow a look at the world simply as it is in itself, free of the vagaries of conceptual description. This checking the facts – those things that are undescribed, ineffable but certain, that is, beyond accountability – calls for an impossible combination. That which is ineffable makes no claim and offers no information, certain or otherwise, that can serve as evidence. The theoretical component that all observations must bear to function as useful, informative, accountable evidence makes observations revisable and even rejectable. They are definitely not right off the answer key. Observations too are questions on the exam. Their justification, like that of the more theoretical claims, will be based on their relations with answers to other questions. They will be evaluated in terms of consistency, explanatory relevance, and relations of entailment with other theoretical and observational reports.

The ambition of science, to describe more than is apparent, keeps it steadily at risk of being wrong. There is always an informational gap between the objects of theories and the evidence used to formulate and verify the theories. Claims about the microworld, the very distant world, or the past are motivated and justified by observations of events and objects that are large, nearby, and present. The accessible data are in features of observable phenomena that are caused by the unobservable objects of theory. The image we see of a DNA molecule as projected on the

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screen of an electron microscope is not the molecule itself, but it is caused by the molecule. Similarly, the tracks photographed in a particle detector or the clicks of a particle counter are not themselves subatomic particles, but they are caused by those particles. The effects bear information of the cause, and the challenge of science is to discover features of the cause through evidence in the effect. Effects are all we have to work with.

Effects are informative of their causes only with the assistance of causal theory. That is, claims about the unobservable causes are products of inference. And the inference must be sanctioned by a theoretical conditional of the form: If the observable effect has feature x , then the unobservable cause has feature y . With evidence and object of theory on opposite ends of this interaction chain, the available data can be useful evidence only in concert with theories to describe the chain and trace the flow of information from one end to another. We are stuck at the effects end of this chain, always interested in the causal end.

However the process of justification in science goes, it must acknowledge this predicament. It must pay attention to what information is available to us and what is in principle beyond the limits of awareness. Only the available information can be relied on to guide the justification and to supply the proof of theories.

JUSTIFICATION AS COHERENCE

Justification is to be a symptom of truth. A properly designed process of justification will allow us to say that if a belief, a theory, is justified then it is likely to be true. The justification will take place either by comparing the theory to the world to see if the description corresponds to the facts or by comparing the theory to other theories to see if it fits into the theoretical system and coheres with the web of theories. The former relation of correspondence with the facts is indeed what we mean by the truth. Correspondence is the nature of truth, but it is exactly this relation

of correspondence that we are unable to evaluate. The world side of this theory—world correspondence is inaccessible information. It is the causal end of the interaction chain. It is the answer key we never get to see and so never should rely on for guiding scientific decisions. Much as we would like to use correspondence-with-the-facts as the measure of justification, we can't. We simply have no direct way of telling when correspondence has been achieved.

We are stuck with coherence, judging theories by their relation to other theories. It is not that coherence *is* truth; rather, coherence *indicates* truth. This claim still needs some work, but a system of coherent theories is more likely to correspond to the way things are in the world than is a less coherent system. Coherence, if it is to be the standard of justification of theories, must indicate correspondence.

One thing is for sure at this point: If there is any way to justify scientific theories it will have to be by coherence, by judging the relation between theories. This is assured on the authority of the prerequisite that justification be evaluated on the basis of available information. The method of justifying theories will involve testing by comparison to other theories and by comparison to theory-sponsored evidence. The accomplishment of justification is finding a high degree of coherence among these factors. Thus the process and result of justification are all internal to one's cognitive awareness. Not that it must be grasped all at once by an individual scientist, but the process must be humanly accessible. As science is a communal undertaking, the process of justification, the standards of responsibility in what to believe, must be internal to the community of investigators. Scientific theories, in other words, must be assessed from within the theoretical system of science. There is just nothing for it; we cannot step outside the system for a peek at the facts. And it is not for moral or social reasons that this is the way things are. It's not a matter of pig-headedness or willful bias. It's just the way things are. We must

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do the best we can with the information that is available, and that means using coherence as the measure of justification and the indicator of truth.

The distinction between internal and external features of theories has been abandoned, and we have settled our attention on the internal side, all for reasons of availability of information. Truly external features are not within our domain of awareness and so are not worth considering when we are looking for useful ways to evaluate theories. The same concern and the same argument force us to abandon the distinction between theories and facts. If facts are the direct information of the way things are in the world, pure and uncolored by influences of description, with no filtering of preconception, then they are beyond our cognitive grasp. If there are such things as facts, they are useless to science because they are unavailable to science. This doesn't mean that there are no such things as facts and that all there are are theories. It means that we cannot know of the facts and so cannot appeal to them in justification. As far as we are aware, there is only theory, and that's as far as we can go in a responsible, serviceable method of justification. The internal–external and theory–fact distinctions are conflated in that anything that is useful to us, anything recognizable and discussible and accountable, is on the internal-theory side. What we take to be facts, observations of the world, are laced with theory.

Denying the fact–theory dichotomy, restricting attention to the theory side of things, is not a suggestion of anarchy. It doesn't lead to a science that is thoroughly subjective, where anything goes and justification is by fiat or caprice. There still are standards, internal standards as discussed in Chapter 2, and there still are internal procedures (more to come on these) to provide guidelines for justification. Though the process must be from within the system of beliefs and theories, it need not be subjective. We can hold out for a standard of objectivity that is complete with the information available, a notion of internal objectivity.

OBJECTIVITY FROM WITHIN

Abandon the internal–external dichotomy and conflate the parallel distinction between theory and fact, but don’t give up on an objective–subjective distinction. Objectivity can be characterized in two ways, and one of them is amenable to entirely internal evaluation.

One way we could use the concept of objectivity in science would be to describe a statement, like the statement of a theory, as being objective if it is accurate about the object it describes. Objective statements are about real things; subjective statements are about things and properties we make up. This makes objectivity a very desirable feature of science in that it removes the judgment from the control of people. Objectivity, in this sense of the statement being in fact about a real, external object, is good because it blocks any influence of whimsy, prejudice, dementia, bias, or any other human persuasions that steer theory away from truth. It is a wonderfully clear and firm and noble standard. It is also utterly useless. It is warmed-over correspondence, and we have no means of evaluating the link between statement and object in the world.

There is another way to employ the notion of objectivity. The idea is still to avoid the human influences and biases, the subjective factors, that are obstacles to the truth. This aim is furthered by an insistence on objective processes and methods in science. Science is done objectively insofar as it is done with an openness to discussion and a willingness to hear and consider objections to any theory. Objectivity in this sense is in the spirit of putting one’s ideas on the line, being open to testing and revision. It is being open-minded and not insular, defending a theory by confronting any challenges rather than by insulating it from challenges. Opening the process of justification in this way removes it from the private and personal influences associated with subjectivity. Requiring an ongoing attitude of openness to challenge iso-

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lates the justification process from the personal or social desires of the scientists. It prevents our believing in a theory simply because we *want* to believe it. And the good news is that this objectivity of process, in contrast to the objectivity of a statement, is something that can be evaluated. From within the system we can tell if a theory is testable and challengeable and thereby amenable to an objective process of justification.

This style of objectivity is closely linked to intersubjectivity. Matters of justification cannot be privately controlled. An open, intersubjective process will constrain the individual's influence. It does not, however, constrain the influence and control of the community, and given that scientists band in rather homogeneous groups of similarly trained individuals, it is not so clear that the vagaries of community decisions won't simply reproduce the vagaries of its individuals' decisions.

This concern helps to point out that the concept of internalism can be broken into levels. There are different levels of internalism in the sense that an intellectual process, such as justification of claims about the world, could be restricted to the information available to a particular person, to the information available to the whole community, or even to every person dead or alive. There is a level of internalism that is internal-to-a-particular-theory, meaning that arguments and evaluations of claims are in terms of concepts and standards that are under the influence of that theory in particular. But there is also a level of internal-to-theories-in-general, which requires that our knowledgeable activities be dependent on information within the ensemble of theories we hold about the world. It is this last, most generous version of internalism that is the predicament of science as sketched above.

Thomas Kuhn, in his influential book, *The Structure of Scientific Revolutions* (first edition, 1962), introduces yet another shade of internalism. Because of the high profile of Kuhn's account of science, it is wise to locate his position with respect to the account

unfolding here. Kuhn is most concerned with the succession of scientific theories, as from Newtonian mechanics to relativity, or from the caloric theory of heat to the kinetic theory, and the assessment of the new theory as being somehow better than the old. The evaluation of a theory, according to Kuhn, is a process that is internal to what he calls a paradigm. This concept of a paradigm is the key to Kuhn's model of science. The current paradigm of a particular science includes the theoretical commitments of the scientific community, that is, the theoretical claims it endorses and the theoretical terms it uses, as well as the experimental procedures and standards accepted as valuable. For example, physics since the 1930s has been done within what could be called the quantum-mechanics paradigm. Virtually the entire community of physicists conduct their investigation of the world from the perspective of the quantum theory, which prescribes what machines to build, what experiments to do, when they have been properly done, and what the results mean. Physicists speak a quantum language and model the world with quantum concepts. It is their mind-set, their world view, their way of speaking and doing business. It is their paradigm.

Since, according to Kuhn, the paradigm in which scientists work influences the meaning and evaluation of the observations they make, the justification of theories, which includes the comparison of theories with evidence, is internal to a paradigm. Theories can be justified in the context of this paradigm or justified in the context of that paradigm, but there is no chance of justification pure and simple. Thus Kuhn's internalism is somewhere between the narrow internalism-to-a-particular-theory and the broad internalism-to-theories-in-general. It is an internalism to a particular ensemble of theories-plus-experimental-guidelines, internalism to a paradigm.

This level of internalism leads to an interesting and controversial description of how scientists decide to abandon one account of the world and adopt a new one. Whole paradigms, in Kuhn's

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model, rather than individual theoretical claims, are the units of scientific change. There are no tests of theories that can be done independently of all paradigms, so each paradigm is left to assess its own theories, sponsor its own evidence, and set its own standards. Alternative paradigms, such as the one that embeds the caloric theory and that of the kinetic theory, do not even use the same language and so cannot really be directly compared. They are likely to specify different experiments as the important tests of a theory of heat and to specify different standards and meanings to observation. The scientific allegiance switches from one to the other, according to Kuhn, not because of some objective testing, that is, testing that is uninfluenced by either paradigm, testing that rules in favor of one theory over the other. It is rather a wholesale switch of paradigms including the theories, experimental standards, important experiments, and all, a switch motivated by an accumulation of unsolved problems within the old paradigm and a willingness to adopt the whole of the new paradigm.

Kuhn's point is that the basis for asserting one account of the world over another is not and cannot be objective in the sense of being founded on brute facts that are independent of either account and therefore agreeable to both parties as adjudicator of the issue. There are no such facts, since all information is internal to a paradigm and hence in collusion with one account or the other.

This reasserts the point that objectivity in the form of brute facts in manifest correspondence to the world is beyond our means. It also sharpens the burden on objectivity as an openness to challenge from a variety of perspectives. The burden is to show that the variety can cross the boundaries of paradigms and allow for meaningful tests that genuinely put the theory at risk.

Objectivity of process, in other words, is a nice opening suggestion for a notion of internal objectivity. It is clearly based on available information, but it falls a bit short of our expectation of all that objectivity ought to be, and it is, at this point, a bit thin

on detail. We will have further suggestions for the internal concept of objectivity and its evaluation in Chapter 9.

This chapter is short because its message is so simple and yet so important. The kernel of the message is a humble truism: Whatever you use to prove your theories, to justify the belief in their truth, must be information that is accessible in the sense of being within the domain of human awareness. We must be aware, or at least potentially aware, of the justifying factors used to establish the credibility of scientific theories and to keep science on a responsible, truth-conducive method. Justifying factors must be recognizable, discussible, and accountable. If they are observations they must be described. Justifying factors, in other words, will be theoretical factors. We will always be at the mercy of theory. The assessment of scientific claims, the justification of theories, is confined to be done from within a theoretical system. Coherence is the only option, so we had better be sure it has something to do with the truth.